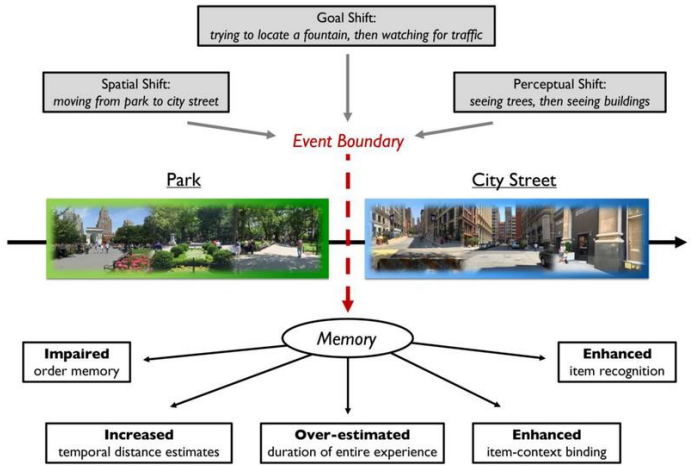
Motivation

- Current video memory pipelines are linear and rely on flat query-frame retrieval for question-answering

- Limits capability for multi-hop question-answering: retrieve and aggregate scattered information across video to answer question
  - retrieval degenerates into selecting events purely by query similarity
  - return scattered set of moments rather than a coherent sequence
  - can miss contextually related events that are relevant but not directly similar to the query.

Motivation

- Human Memory:
  - Learns to separate continuous experience into discrete "events"
    - Intrinsic contexts: goals, actions
    - Extrinsic contexts: location, spatial surroundings
  - Able to link events together based on these shared contexts



Key Research Questions

1. Is graph-based memory parsing better than raw memory retrieval? Does the cross-memory relation benefit understanding?

2. Is it possible to dynamically change the meaning of memory nodes based on what the user is looking for?
    1. Is there a broad manifold for general-purpose understanding with learnable transformations to adapt it into a manifold for task-speciifc understanding

Proposed Idea

- Build memory as a structured event graph, linking dynamic sets of frames (events) based on shared context
  - “Events” is an overarching term here implying objects, scenes, actions, etc.

- Query-Adaptive Traversal and Retrieval
  - The constructed graph should be dynamic, such that edge weights are modulated by the query.
  - Examples:
    - How did the man's appearance change from the beginning of the day to the end?
    - When were the man and the dog together?
    - What happened to the man after the fire alarm went off?